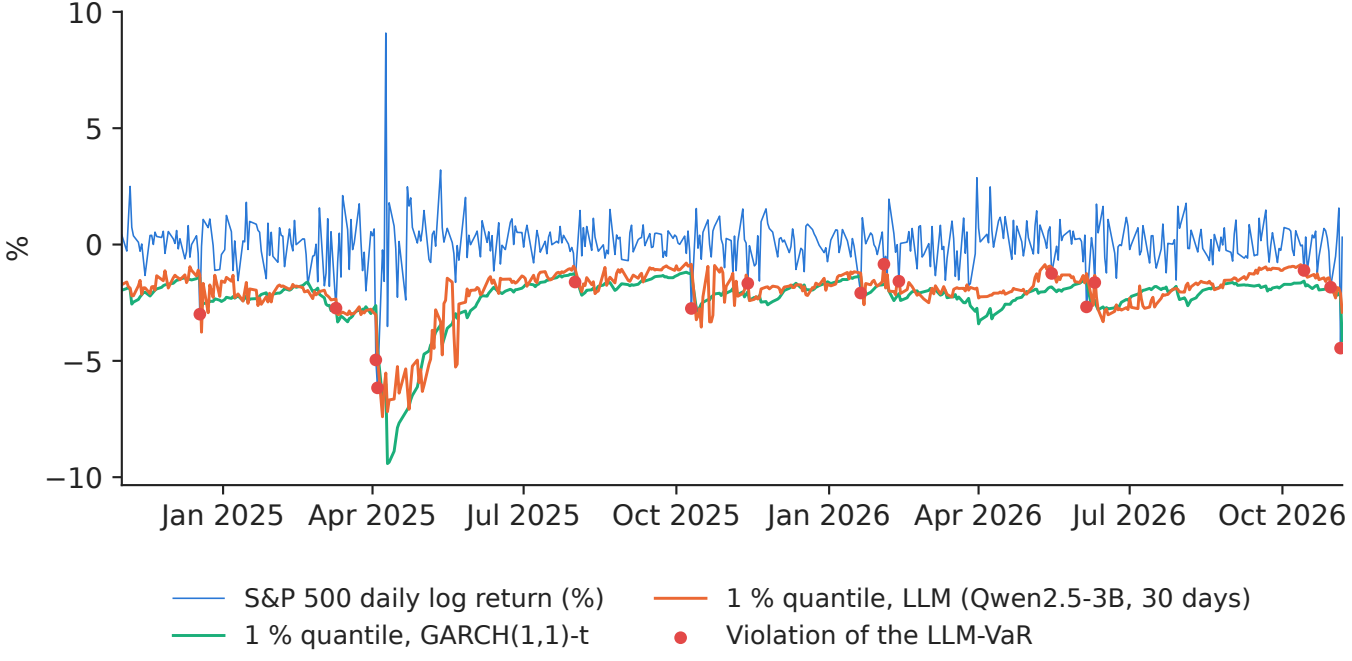
10
5
0
−5
−10
%
Jan 2025
Apr 2025
Jul 2025
Oct 2025
Jan 2026
Apr 2026
Jul 2026
Oct 2026
S&P 500 daily log return (%)
1 % quantile, GARCH(1,1)-t
1 % quantile, LLM (Qwen2.5-3B, 30 days)
Violation of the LLM-VaR